THE NUMBER OF GEOMETRIC BRANCHES AT A POINT IS INVARIANT UNDER EXTENSION OF THE BASE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains a commutative diagram typeset with Xy-pic; the retained chapter preamble supplies the native Xy-pic setup, so the diagram compiles exactly as in the source.

Editorial note. The counted span cites six cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The final algebra fact is attributed by the author to an elementary argument as well as to the cited lemma; the counted span contains the author's own reduction to that fact.

Allowed standard prerequisites (editorial).

- Normalization commutes with field extensions up to universal homeomorphism. Stacks Project, Varieties, Lemma lemma-normalization-and-change-of-fields (tag 0C3N). Identifies the top horizontal arrow of the diagram and shows it is a universal homeomorphism.
- Geometric branches as separable degree sums and connected components over separably closed fields. Stacks Project, Varieties, Lemma lemma-number-of-branches (tag 0C1S) and Lemma lemma-separably-closed-field-connected-components (tag 0363). The first gives the branch count along the normalization and the second converts the algebra fact into a connected-component count.

Source reference. Original lemma 0C3N: sources/stacks-chow/tags.txt, source lines 15198–15198. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04DF: sources/stacks-chow/tags.txt, source lines 5725–5725. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C1S: sources/stacks-chow/tags.txt, source lines 15132–15132. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JT: sources/stacks-chow/tags.txt, source lines 2026–2026. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0363: sources/stacks-chow/tags.txt, source lines 4150–4150. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme. Let K/k be an extension of fields. Let $y \in X_K$ be a point with image x in X. Then the number of geometric branches of X at x is the number of geometric branches of X_K at y.*

Proof. Write $Y = X_K$ and let X^ν, resp. Y^ν be the normalization of X, resp. Y. Consider the commutative diagram

$$\begin{array}{ccccc} Y^\nu & \longrightarrow & X^\nu_K & \longrightarrow & X^\nu \\ \downarrow & & \downarrow{\scriptstyle \nu_K} & & \downarrow{\scriptstyle \nu} \\ Y & = & Y & \longrightarrow & X \end{array}$$

By Lemma 0C3N we see that the left top horizontal arrow is a universal homeomorphism. Hence it induces purely inseparable residue field extensions, see Morphisms, Lemmas 04DF and 01S4. Thus the number of geometric branches of Y at y is $\sum_{\nu_K(y')=y}[\kappa(y') : \kappa(y)]_s$ by Lemma 0C1S. Similarly $\sum_{\nu(x')=x}[\kappa(x') : \kappa(x)]_s$ is the number of geometric branches of X at x. Using Schemes, Lemma 01JT our statement follows from the following algebra fact: given a field extension l/κ and an algebraic field extension m/κ, then

$$\sum\nolimits_{m\otimes_\kappa l \to m'} [m' : l']_s = [m : \kappa]_s$$

where the sum is over the quotient fields of $m \otimes_\kappa l$. One can prove this in an elementary way, or one can use Lemma 0363 applied to

$$\mathrm{Spec}(m\otimes_\kappa l)\times_{\mathrm{Spec}(l)}\mathrm{Spec}(\bar{l}) = \mathrm{Spec}(m)\otimes_{\mathrm{Spec}(\kappa)}\mathrm{Spec}(\bar{l}) \longrightarrow \mathrm{Spec}(m)\times_{\mathrm{Spec}(\kappa)}\mathrm{Spec}(\overline{\kappa})$$

because one can interpret $[m : \kappa]_s$ as the number of connected components of the right hand side and the sum $\sum_{m\otimes_\kappa l\to m'}[m' : l']_s$ as the number of connected components of the left hand side. □